Physics 711 January 22, 2010

Problem Set 9

Problem 1: Exercise 7.4.2

$$\begin{aligned}\langle X\rangle &= \langle n|X|n\rangle = \left(\frac{\hbar}{2m\omega}\right)^{1/2}\langle n|(a^\dagger + a)|n\rangle = \left(\frac{\hbar}{2m\omega}\right)^{1/2}\langle n|\left(\sqrt{n+1}|n+1\rangle + \sqrt{n}|n-1\rangle\right) \\ &= \left(\frac{\hbar}{2m\omega}\right)^{1/2}\left(\sqrt{n+1}\langle n|n+1\rangle + \sqrt{n}\langle n|n-1\rangle\right) = 0,\end{aligned}$$

since $\langle n|n-1\rangle = \langle n|n+1\rangle = 0$ by orthogonality of the energy eigenstates. Similarly,

$$\begin{aligned}\langle P\rangle &= \langle n|P|n\rangle = i\left(\frac{m\omega\hbar}{2}\right)^{1/2}\langle n|(a^\dagger - a)|n\rangle \propto \langle n|\left(\sqrt{n+1}|n+1\rangle - \sqrt{n}|n-1\rangle\right) \\ &\propto \sqrt{n+1}\langle n|n+1\rangle - \sqrt{n}\langle n|n-1\rangle = 0.\end{aligned}$$

Therefore

$$\begin{aligned}\Delta X^2 &= \langle n|X^2|n\rangle = \frac{\hbar}{2m\omega}\langle n|(a^\dagger + a)^2|n\rangle = \frac{\hbar}{2m\omega}\langle n|\left(a^2 + aa^\dagger + a^\dagger a + (a^\dagger)^2\right)|n\rangle \\ &= \frac{\hbar}{2m\omega}\left(\langle n|a\sqrt{n+1}|n+1\rangle + \langle n|a^\dagger\sqrt{n}|n-1\rangle\right) \\ &= \frac{\hbar}{2m\omega}\left(\sqrt{n+1}\sqrt{n+1}\langle n|n\rangle + \sqrt{n}\sqrt{n}\langle n|n\rangle\right) = \frac{(2n+1)\hbar}{2m\omega},\end{aligned}$$

where I dropped the a^2 and $(a^\dagger)^2$ terms in the first line since $a^2|n\rangle \propto |n-2\rangle$ and $(a^\dagger)^2|n\rangle \propto |n+2\rangle$ which are orthogonal to $|n\rangle$, and in the last step I used the normalization condition that $\langle n|n\rangle = 1$. Similarly,

$$\begin{aligned}\Delta P^2 &= \langle n|P^2|n\rangle = -\frac{m\omega\hbar}{2}\langle n|(a^\dagger - a)^2|n\rangle = -\frac{m\omega\hbar}{2}\langle n|\left(a^2 - aa^\dagger - a^\dagger a + (a^\dagger)^2\right)|n\rangle \\ &= \frac{m\omega\hbar}{2}\langle n|(aa^\dagger + a^\dagger a)|n\rangle = \frac{(2n+1)m\omega\hbar}{2},\end{aligned}$$

from the ΔX^2 calculation. Therefore

$$\Delta X\Delta P = \sqrt{2n+1}\left(\frac{\hbar}{2m\omega}\right)^{1/2}\cdot\sqrt{2n+1}\left(\frac{m\omega\hbar}{2}\right)^{1/2} = (2n+1)\frac{\hbar}{2}.$$

Problem 2: Exercise 7.4.3

First the classical result. In a circular orbit of radius r, the acceleration is v^2/r directed towards the center. If the potential is $V = ar^k$, the force directed towards the center is $F = dV/dr = akr^{k-1}$. Thus $F = ma$ implies $akr^{k-1} = mv^2/r$, which can be rewritten as $mv^2/2 = (k/2)ar^k$ which we recognize as the expression $\bar{T} = (k/2)\bar{V}$ where T and V are the kinetic and potential energies of the particle. Here the bars denote the average value, which for circular orbits is redundant since T and V are constant. However, it can be shown that the same averaged relation holds even for non-circular orbits.

For the 1-dimensional harmonic oscillator (where there are no circular orbits!) $V \propto x^2$, so $k = 2$, so we expect $\bar{T} = \bar{V}$. We can check this quantum mechanically:

$$\begin{aligned}
\langle T \rangle &= \langle n|T|n\rangle = \frac{1}{2m}\langle n|P^2|n\rangle = \frac{1}{2m}\frac{m\omega\hbar}{2}(2n+1) = \frac{\omega\hbar}{4}(2n+1), \\
\langle V \rangle &= \langle n|V|n\rangle = \frac{m\omega^2}{2}\langle n|X^2|n\rangle = \frac{m\omega^2}{2}\frac{\hbar}{2m\omega}(2n+1) = \frac{\omega\hbar}{4}(2n+1),
\end{aligned}$$

giving the expected agreement, where we used the results of the previous problem.

Problem 3: Exercise 7.4.8

(1) $L_x = YP_z - ZP_y$, *etc.* No ordering ambiguities arise because only $[X, P_x] = [Y, P_y] = [Z, P_z] = i\hbar$ are non-zero.

(2) Recall that the only non-vanishing Poisson brackets are $\{x, p_x\} = \{y, p_y\} = \{z, p_z\} = 1$. So

$$\begin{aligned}
\{\ell_x, \ell_y\} &= \{\, yp_z - zp_y \,,\, zp_x - xp_z \,\} \\
&= \{\, yp_z \,,\, zp_x \,\} - \{\, yp_z \,,\, xp_z \,\} - \{\, zp_y \,,\, zp_x \,\} + \{\, zp_y \,,\, xp_z \,\} \\
&= yp_x \,\{\, p_z \,,\, z \,\} - 0 - 0 + xp_y \,\{\, z \,,\, p_z \,\} \qquad (1)
\end{aligned}$$

since there are no other non-vanishing Poisson brackets in these expressions. So

$$\{\ell_x, \ell_y\} = -yp_x + xp_y = \ell_z.$$

(3) All the steps in (1) go through unchanged if you just replace $\{,\} \rightarrow [,]$ and lowercase by uppercase, giving

$$[L_x, L_y] = YP_x\,[P_z, Z] + XP_y\,[Z, P_z].$$

Since $[Z, P_z] = i\hbar$, we get

$$[L_x, L_y] = i\hbar(XP_y - YP_x) = i\hbar L_z.$$

Problem 4: Exercise 7.5.4

(1) $\bar{E} \equiv \sum_i E(i)\,P(i) = \sum_i E(i)e^{-\beta E(i)}/Z = (1/Z)\sum_i E(i)e^{-\beta E(i)} = -(1/Z)\sum_i (d/d\beta)(e^{-\beta E(i)}) = -(1/Z)(d/d\beta)(\sum_i e^{-\beta E(i)}) = -(1/Z)(dZ/d\beta) = -d(\ln Z)/d\beta$.

(2) $Z_{\rm cl} = \sum_i e^{-\beta E(i)}$ becomes, upon replacing the sum over states by an integral over phase space,

$$\begin{aligned} Z_{\rm cl} &= \int dx dp\, e^{-\beta E(x,p)} = \int dx dp \,\exp\left\{-\beta\left(\frac{p^2}{2m}+\frac{m\omega^2 x^2}{2}\right)\right\} \\ &= \left(\int_{-\infty}^{\infty} dx\, e^{-\beta m\omega^2 x^2/2}\right)\left(\int_{-\infty}^{\infty} dp\, e^{-\beta p^2/(2m)}\right) = \left(\frac{2\pi}{\beta m\omega^2}\right)^{1/2}\left(\frac{2\pi m}{\beta}\right)^{1/2} = \frac{2\pi}{\omega\beta}. \end{aligned}$$

So, $\bar{E}_{\rm cl} = -d(\ln Z_{\rm cl})/d\beta = -(d/d\beta)\ln(2\pi/\omega\beta) = (d/d\beta)[\ln\beta - \ln(2\pi/\omega)] = 1/\beta$.

(3) $Z_{\rm qu} = \sum_i e^{-\beta E(i)}$ becomes, upon replacing the sum by a sum over the energy eigenbasis,

$$Z_{\rm qu} = \sum_n e^{-\beta E_n} = \sum_{n=0}^{\infty} e^{-\beta(n+\frac{1}{2})\hbar\omega} = e^{-\beta\hbar\omega/2}\sum_{n=0}^{\infty}\left(e^{-\beta\hbar\omega}\right)^n = \frac{e^{-\beta\hbar\omega/2}}{1-e^{-\beta\hbar\omega}}.$$

Thus

$$\begin{aligned} \bar{E}_{\rm qu} &= -\frac{d\ln Z_{\rm qu}}{d\beta} = -\frac{1}{Z_{\rm qu}}\frac{dZ_{\rm qu}}{d\beta} = \frac{e^{-\beta\hbar\omega}-1}{e^{-\beta\hbar\omega/2}}\frac{d}{d\beta}\left(\frac{e^{-\beta\hbar\omega/2}}{e^{-\beta\hbar\omega}-1}\right) \\ &= \frac{e^{-\beta\hbar\omega}-1}{e^{-\beta\hbar\omega/2}}\left(\frac{-(\hbar\omega/2)e^{-\beta\hbar\omega/2}(1-e^{-\beta\hbar\omega}) - e^{-\beta\hbar\omega/2}(\hbar\omega)e^{-\beta\hbar\omega}}{(1-e^{-\beta\hbar\omega})^2}\right) \\ &= \hbar\omega\left(\frac{1}{2}+\frac{1}{e^{\beta\hbar\omega}-1}\right). \end{aligned}$$

(4) From the above expression

$$\lim_{T\to\infty}\bar{E}_{\rm qu} = \lim_{\beta\to 0}\bar{E}_{\rm qu} = \hbar\omega\left(\frac{1}{2}+\frac{1}{\lim_{\beta\to 0}(e^{\beta\hbar\omega}-1)}\right).$$

If $\beta\hbar\omega \ll 1$, then $e^{\beta\hbar\omega} \approx 1+\beta\hbar\omega+\mathcal{O}(\beta\hbar\omega)^2$, so $\lim_{\beta\to 0}(e^{\beta\hbar\omega}-1) \approx \beta\hbar\omega$ and therefore

$$\lim_{T\to\infty}\bar{E}_{\rm qu} = \lim_{\beta\to 0}\ \hbar\omega\left(\frac{1}{2}+\frac{1}{\beta\hbar\omega}\right) \approx \frac{1}{\beta} = \bar{E}_{\rm cl}.$$

(5) Using the result from part (2) that for each of the $3N_0$ modes (which I'll label by an index $i = 1,\ldots,3N_0$) $\bar{E}_{\rm cl}(i) = \beta^{-1} = kT$,

$$C_{\rm cl} \equiv \frac{1}{N_0}\frac{d\bar{E}_{\rm cl}}{dT} = \frac{1}{N_0}\frac{d}{dT}\sum_{i=1}^{3N_0}\bar{E}_{\rm cl}(i) = \frac{1}{N_0}\frac{d}{dT}\sum_{i=1}^{3N_0} kT = \frac{1}{N_0}3N_0 k = 3k.$$

Quantum mechanically, using the result from part (3) for each mode gives

$$\begin{aligned}
C_{\rm qu} &\equiv \frac{1}{N_0}\frac{d\bar{E}_{\rm qu}}{dT} = \frac{1}{N_0}\frac{d}{dT}\sum_{i=1}^{3N_0}\hbar\omega\left(\frac{1}{2}+\frac{1}{e^{\beta\hbar\omega}-1}\right)\\
&= \frac{3N_0\hbar\omega}{N_0}\frac{d}{dT}\left(\frac{1}{2}+\frac{1}{e^{\theta_E/T}-1}\right) = \frac{-3\hbar\omega}{(e^{\theta_E/T}-1)^2}\frac{d}{dT}(e^{\theta_E/T}-1)\\
&= \frac{-3\hbar\omega}{(e^{\theta_E/T}-1)^2}\left(\frac{-\theta_E}{T^2}\right)e^{\theta_E/T} = 3k\left(\frac{\theta_E}{T}\right)^2\frac{e^{\theta_E/T}}{(e^{\theta_E/T}-1)^2},
\end{aligned}$$

where in the second line we have written $\beta\hbar\omega = \hbar\omega/(kT) \equiv \theta_E/T$.
For $T \gg \theta_E$, $e^{\theta_E/T} \approx 1 + (\theta_E/T) + \cdots$, so

$$C_{\rm qu} \approx 3k\left(\frac{\theta_E}{T}\right)^2\frac{1+\frac{\theta_E}{T}+\cdots}{(\frac{\theta_E}{T}+\cdots)^2} \approx 3k.$$

For $T \ll \theta_E$, $e^{\theta_E/T} \gg 1$, so $e^{\theta_E/T}-1 \approx e^{\theta_E/T}$ and

$$C_{\rm qu} \approx 3k\left(\frac{\theta_E}{T}\right)^2\frac{e^{\theta_E/T}}{(e^{\theta_E/T})^2} = 3k\left(\frac{\theta_E}{T}\right)^2 e^{-\theta_E/T}.$$